

Rotor and Wake Aerodynamics

Short introduction to Actuator Surface and Momentum theory

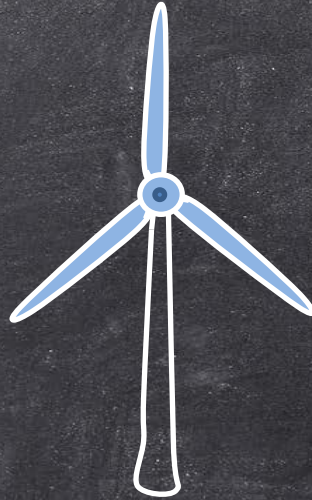


Learning objectives

- To understand the concept of an Actuator Surface,
 - a distribution of forces in space that exchanges momentum and energy with the flow, and accelerates the flow
 - The surface can also be a volume, and can change in time
- To understand that rotors are only devices that approximate a desired force distribution, by sweeping a surface in space and time

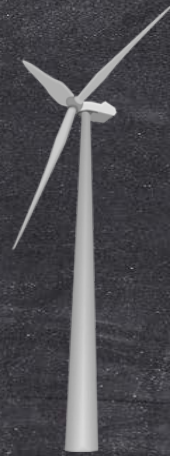
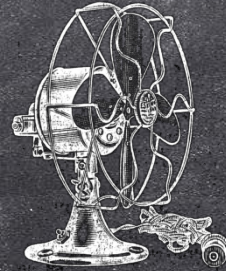


How do we build a model for rotors?



Why do we want to model rotors?

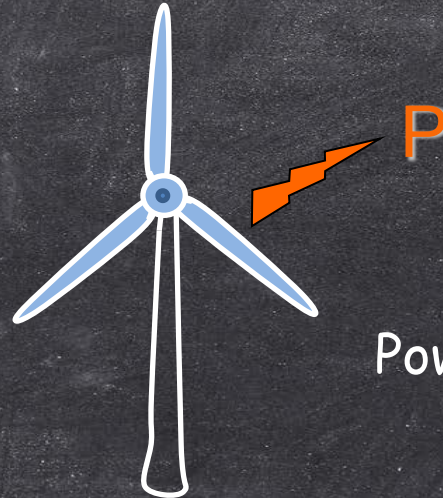
- To make wind flow
- To create propulsion
- To convert power



Wind turbine

The wind applies
torque to the
rotor

The kinetic energy
of the wind is
converted into
mechanical energy

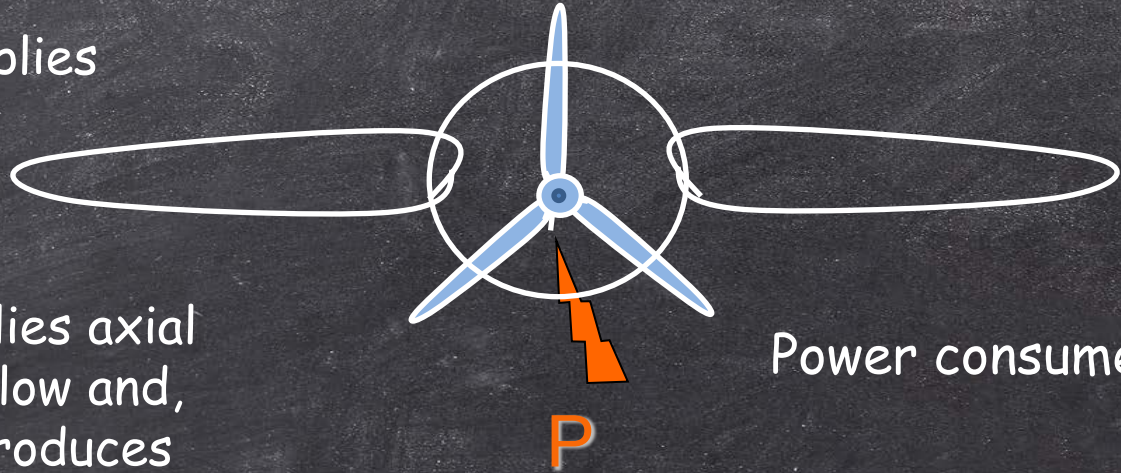


Power extracted

Propeller

The engine applies torque to the rotor

The rotor applies axial force to the flow and, by reaction, produces thrust $\rightarrow$ propulsion



Power consumed

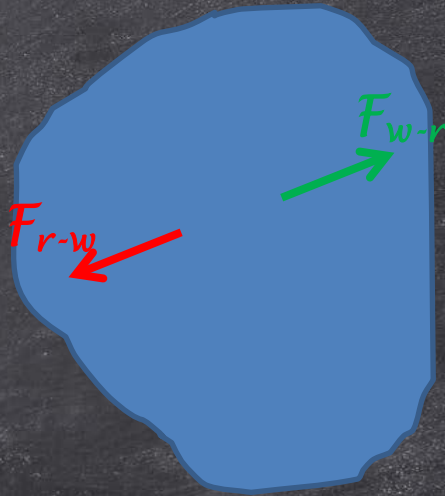
The rotating blades describe a surface of actuation where loads are applied



To design a rotor, we first design a surface or a volume distribution of force

- It can be 2D or 3D

Basic principle:
Newton's 3rd Law!



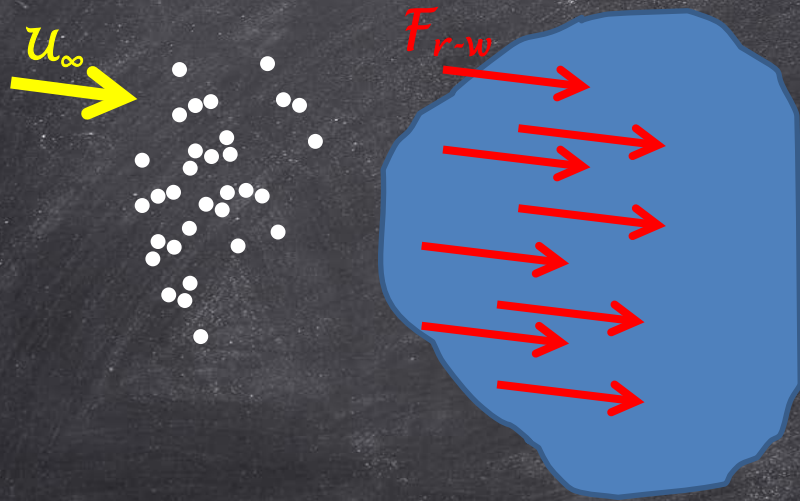
F_{r-w} Force applied to the fluid

F_{w-r} Force applied to the surface by the flow



As unconstrained fluid goes through this volume of forces, it is accelerated or decelerated

Power is converted between the flow and the surface or volume



When the forces on the surface are aligned with the flow, total energy increases

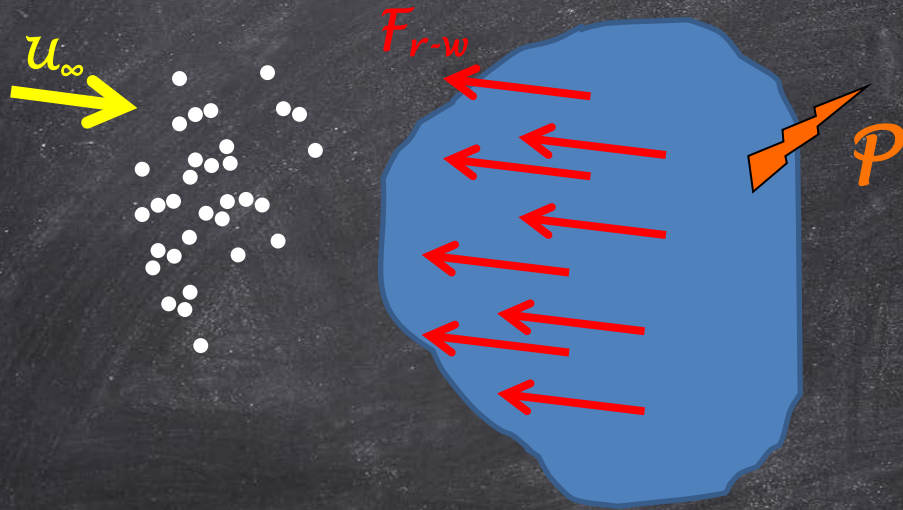


Kinetic energy increases

Power provided to the flow

Contraction

When the forces on the surface are against the flow, total energy decreases



Kinetic energy decreases

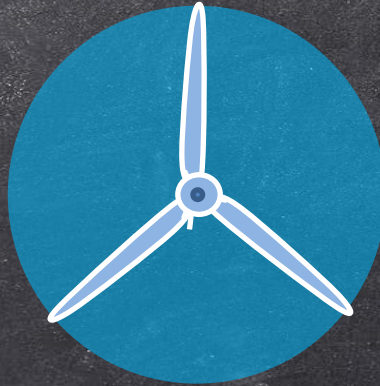
Power extracted from the flow

Expansion

We use rotors to materialise these force distributions over space



One of the most common actuator surfaces is
the actuator disk



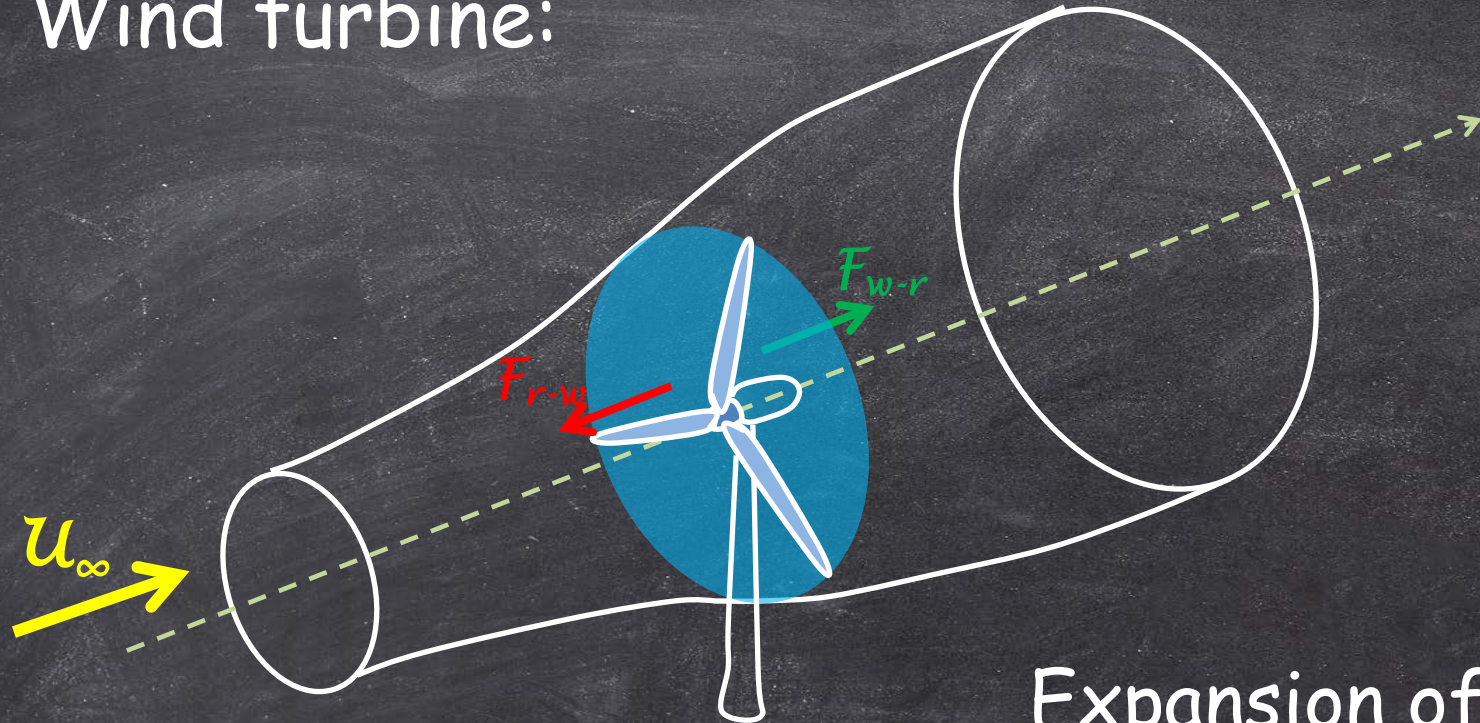
Different rotors materialize different surfaces of forces



- Exercise, imagine the shape of the surfaces generated

Action of the forces on the flow

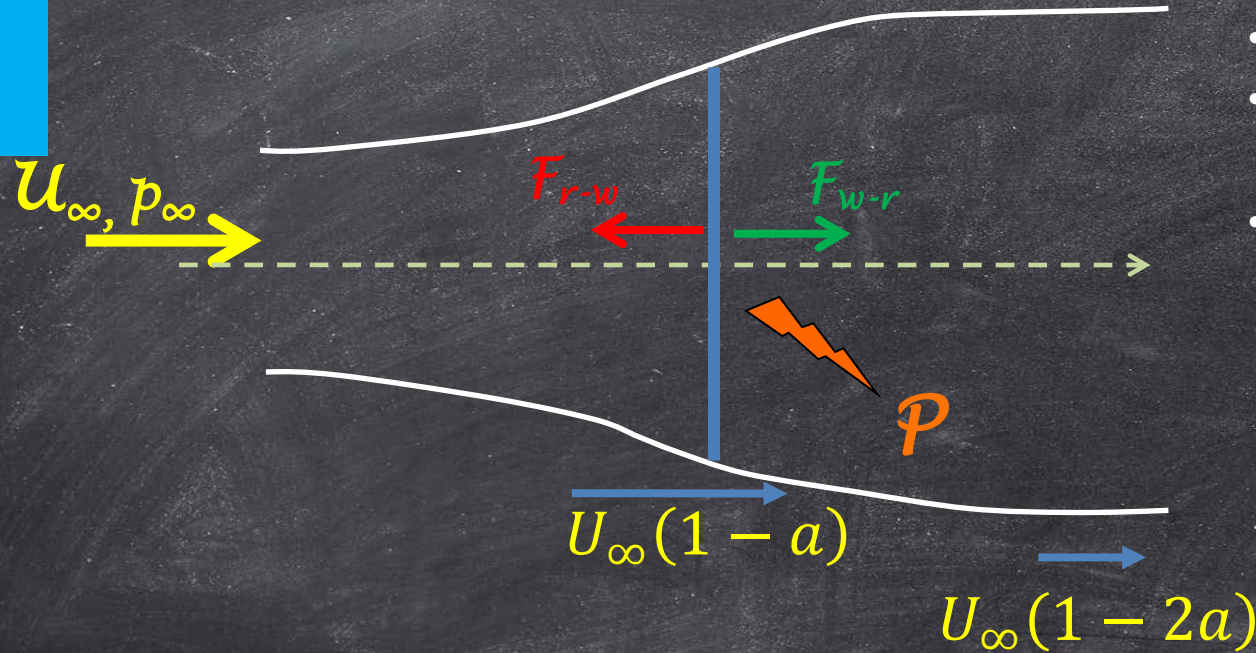
Wind turbine:



Expansion of the
wake



Horizontal-axis Wind turbine:



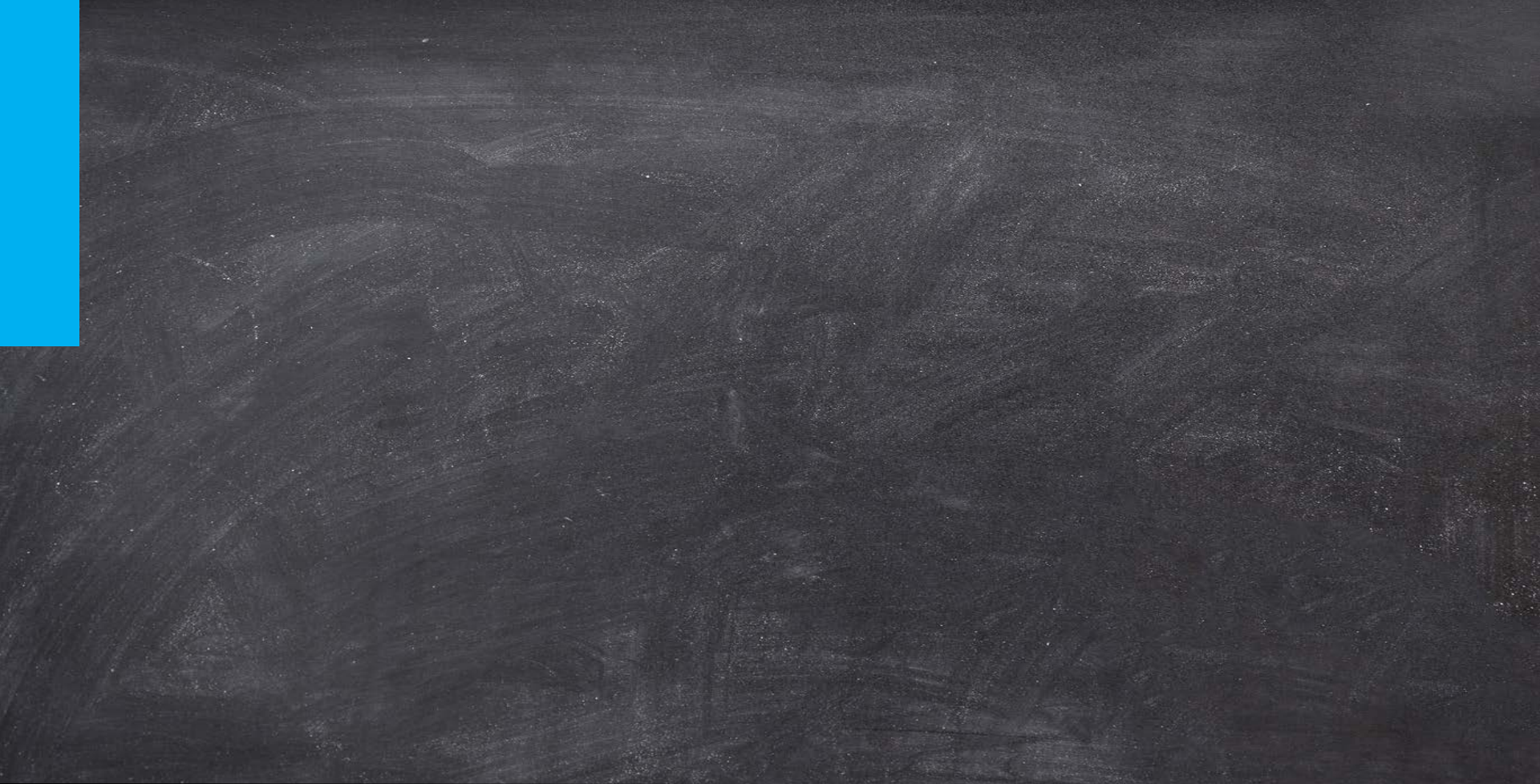
- The wake expands
- The wind speed decreases (negative induced velocity)
- Power extracted from the flow

$$P = U_\infty \cdot F_{r-w}$$

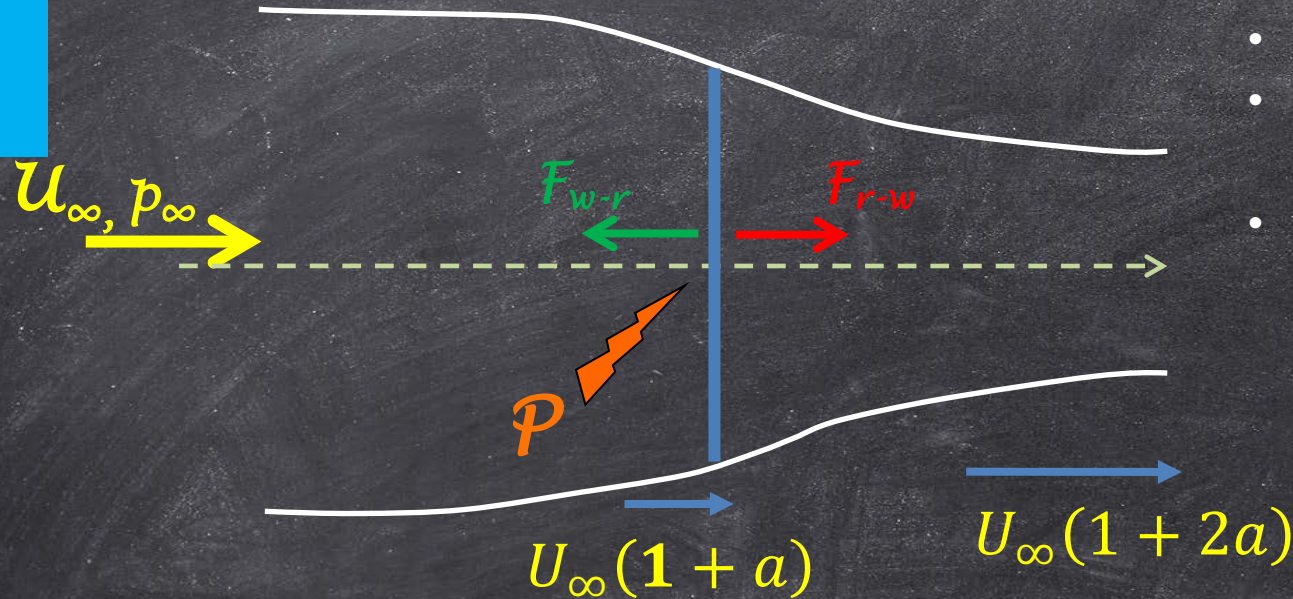
<0 >0 <0

$a = \text{induction factor}$





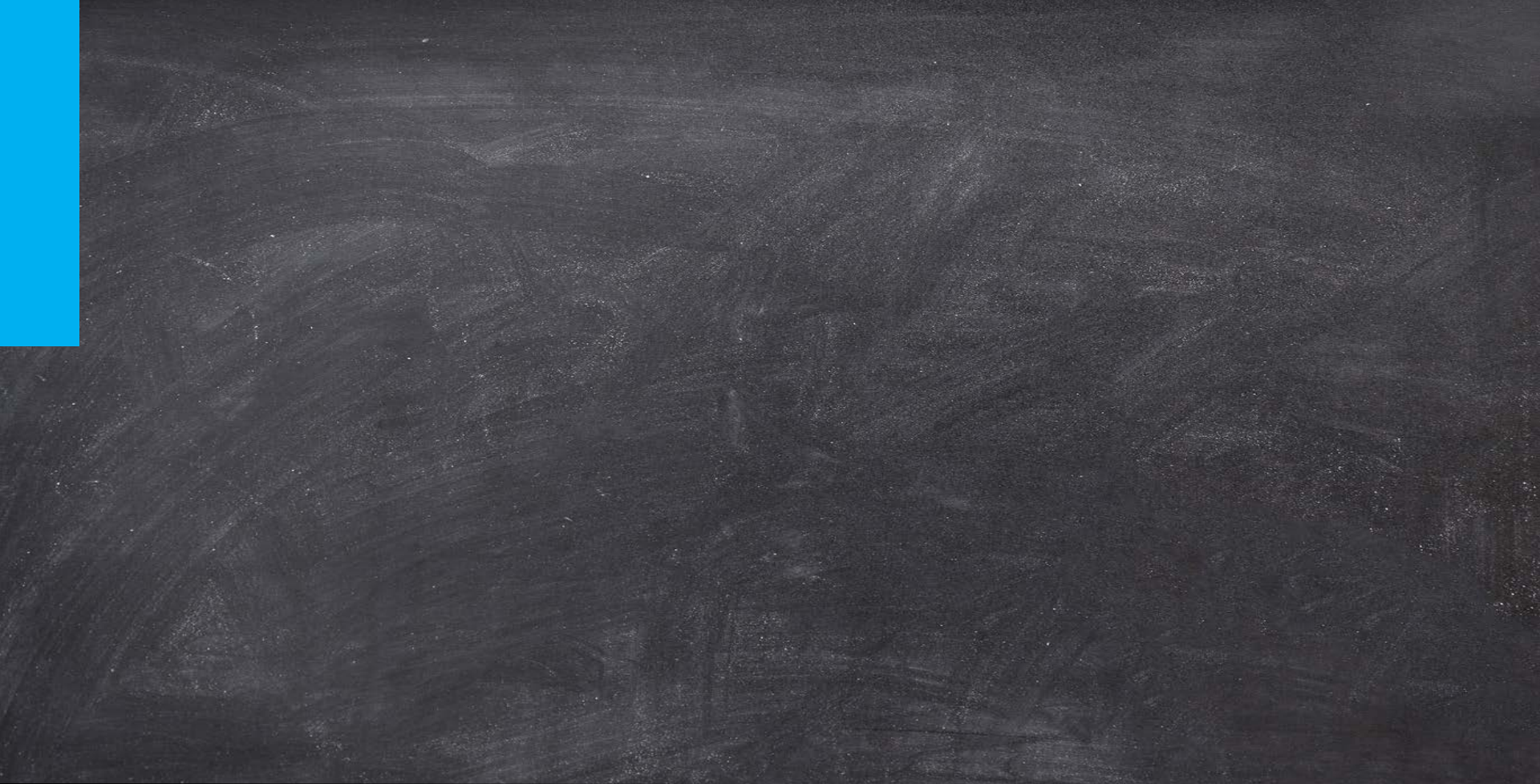
Propellers and fans



- The wake contracts
- The wind speed increases (positive induced velocity)
- Power delivered to the flow

$$P = U_\infty \cdot F_{r-w}$$

$>0 \quad >0 \quad >0$



What we have achieved:

- We understood rotors as mechanisms to generate actuation surfaces
- We understood that the changes of properties of the flow (speed, pressure, energy) are a result of the action of the forces distributed over these surfaces

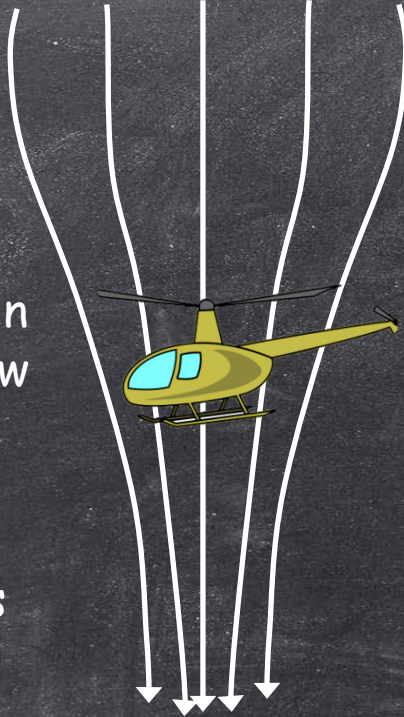
Rotor and Wake Aerodynamics

Rotor wake states



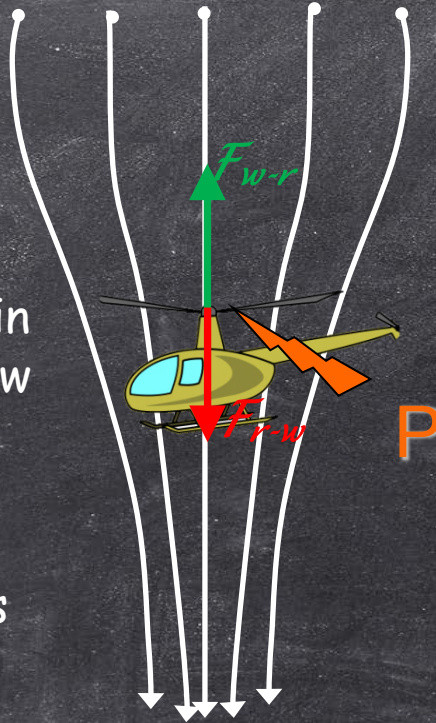
Helicopter state

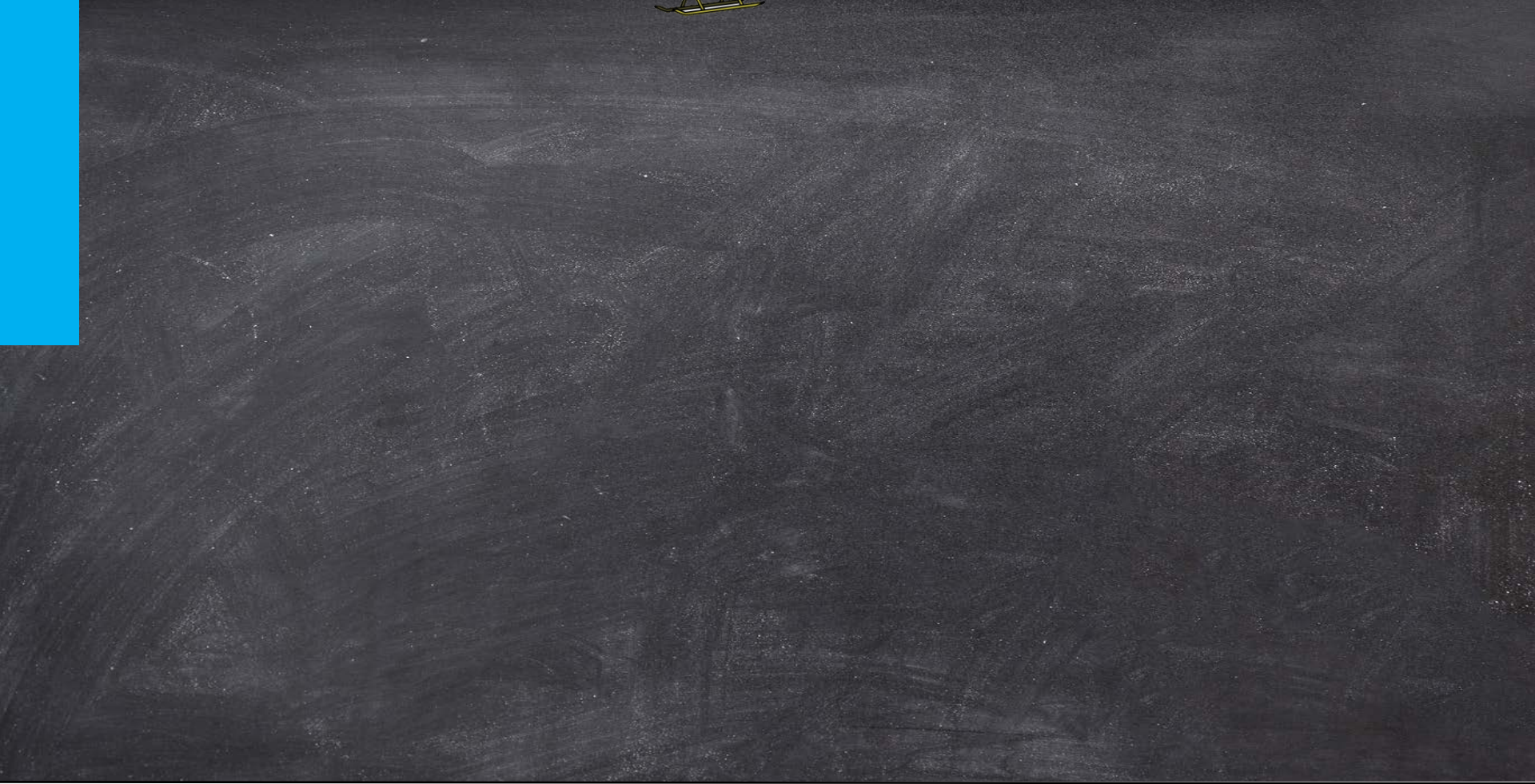
- Thrust force F_{w-r} on the rotor in the opposite direction of the flow
- Energy given to the flow
- Smooth slipstream boundaries - no flow disturbances
- From hover to climb operations in helicopters



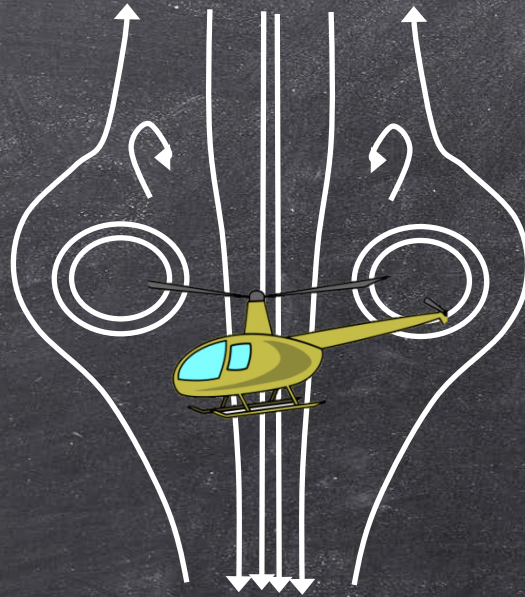
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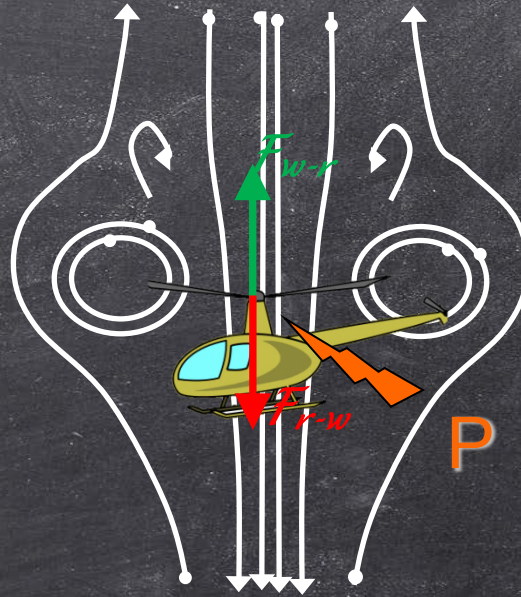


Vortex ring state

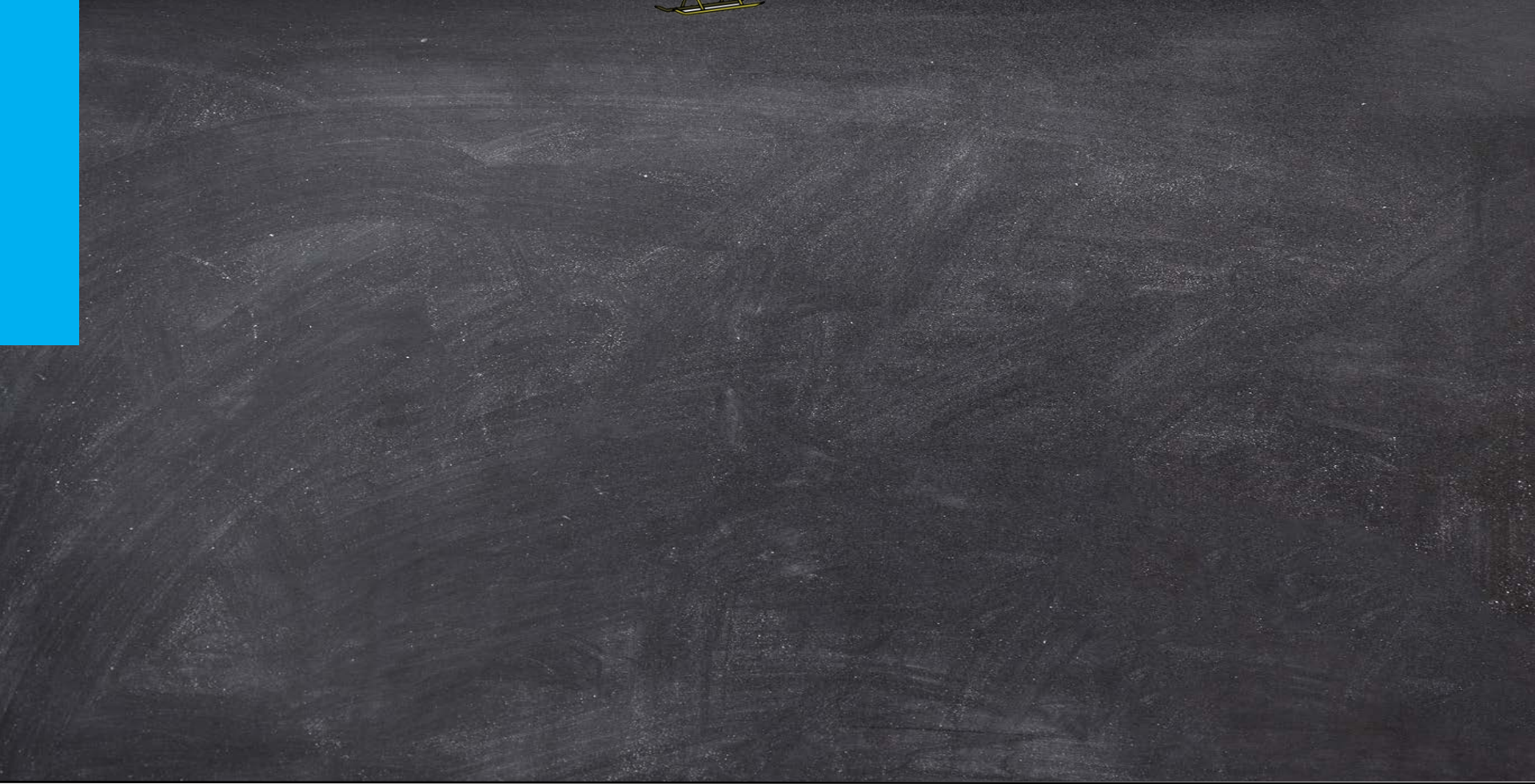


- Thrust force F_{w-r} on the rotor in the opposite direction of the flow
- Energy given to the flow
- Tip vortices close to the rotor form one single structure
- Low descent rate in helicopters

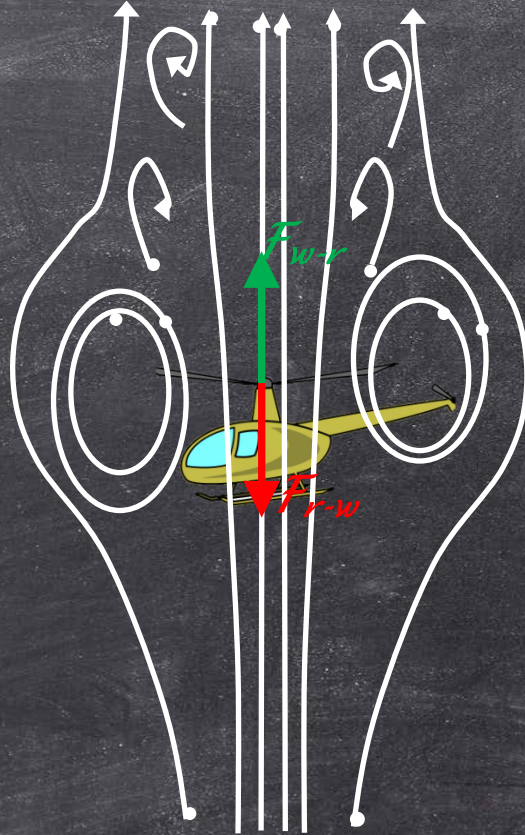
Vortex ring state



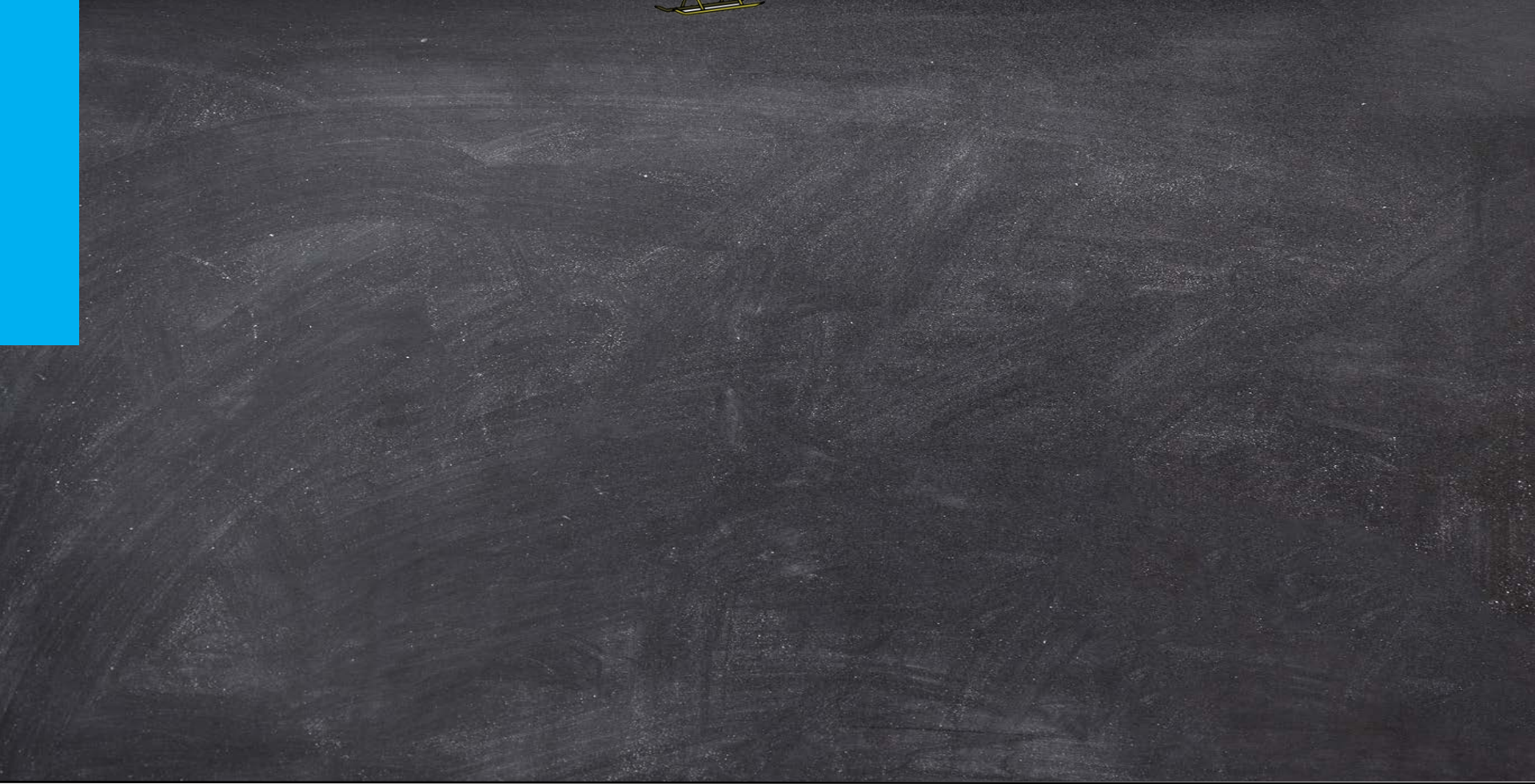
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- Tip vortices close to the rotor form one single structure
- Low descent rate in helicopters



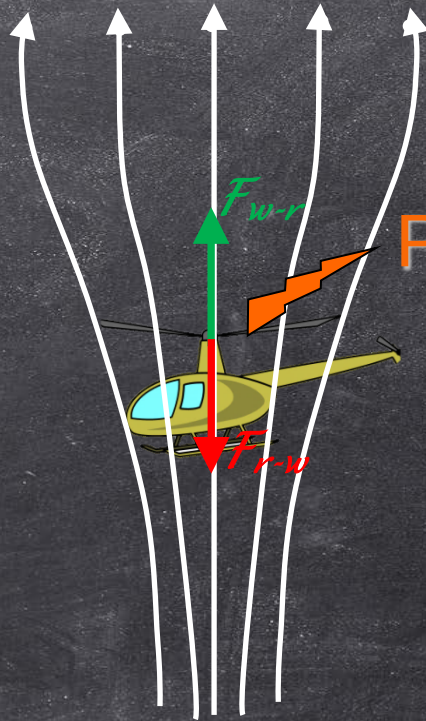
Turbulent wake state



- Thrust force F_{w-r} on the rotor in the same direction of the flow
- Wake above the rotor become turbulent
- Higher descent rate in helicopters
- Transition state, almost no power needed



Windmill state



- Thrust force F_{w-r} on the rotor in the same direction of the flow
- Energy extracted from the flow
- Smooth slipstream boundaries - no flow disturbances
- Normal working operation for wind turbines or very high descent velocity for helicopters
- Power extracted from the flow

Related videos:

- <https://www.youtube.com/watch?v=HrsGMOPzQFo>

Rotor and Wake Aerodynamics

From actuator disc to rotor blade and
airfoil

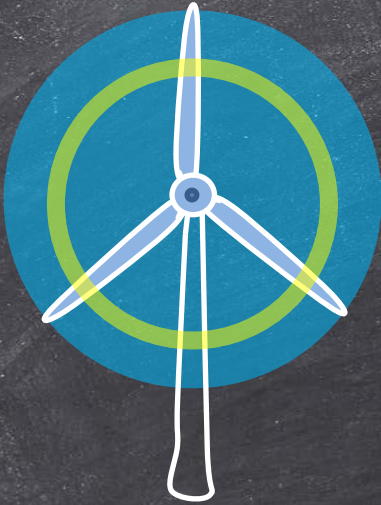
In the previous sub-unit:

- We described the actuator disk model
- Pressure, velocity and loading are uniform at each section of the streamtube

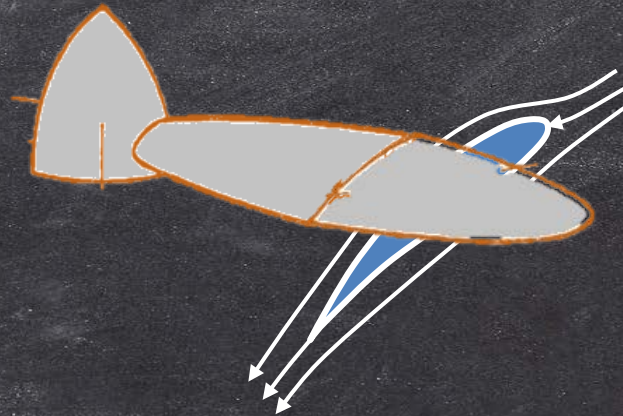
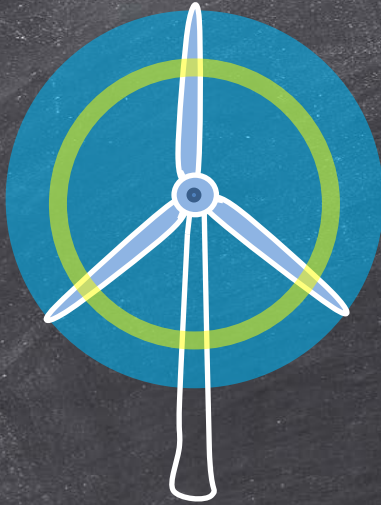
In this sub-unit:

- Pressure, velocity and loading are no longer uniform over each section of the streamtube
- We relate the loading in the streamtube to the loading on the section of the rotor's blade which crosses the streamtube

We will now divide the disk in several annuli

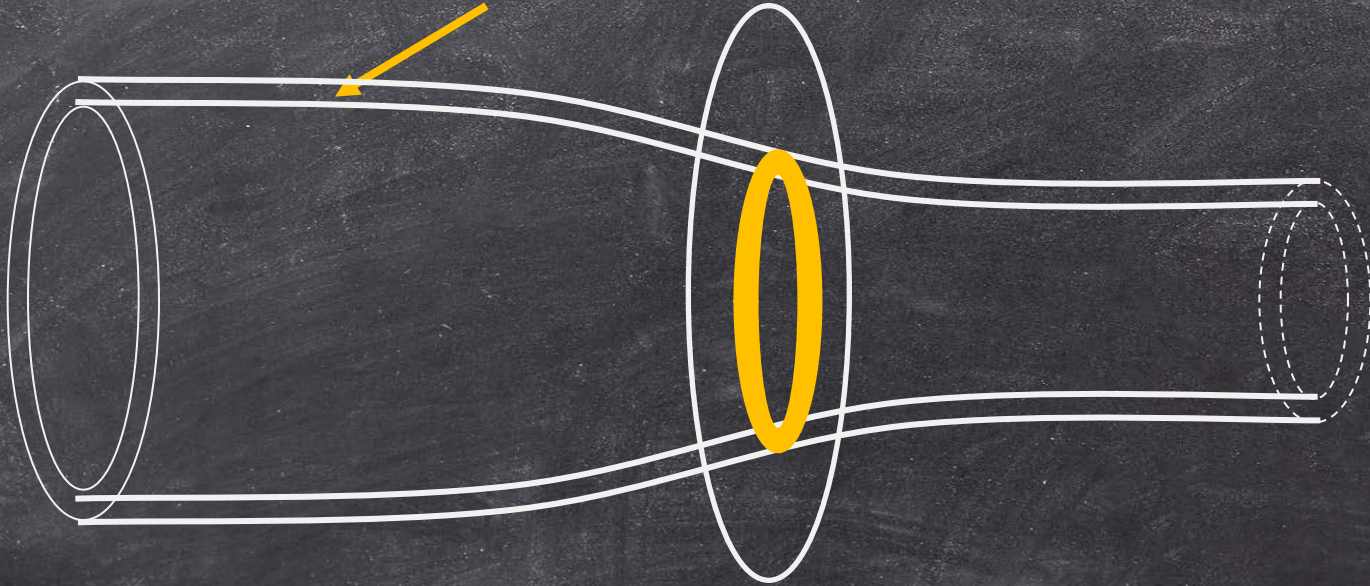


We will now divide the rotor in blades and airfoil section



We apply momentum theory
to each annular streamtube

annular stream tube



We apply momentum theory
to each annular streamtube

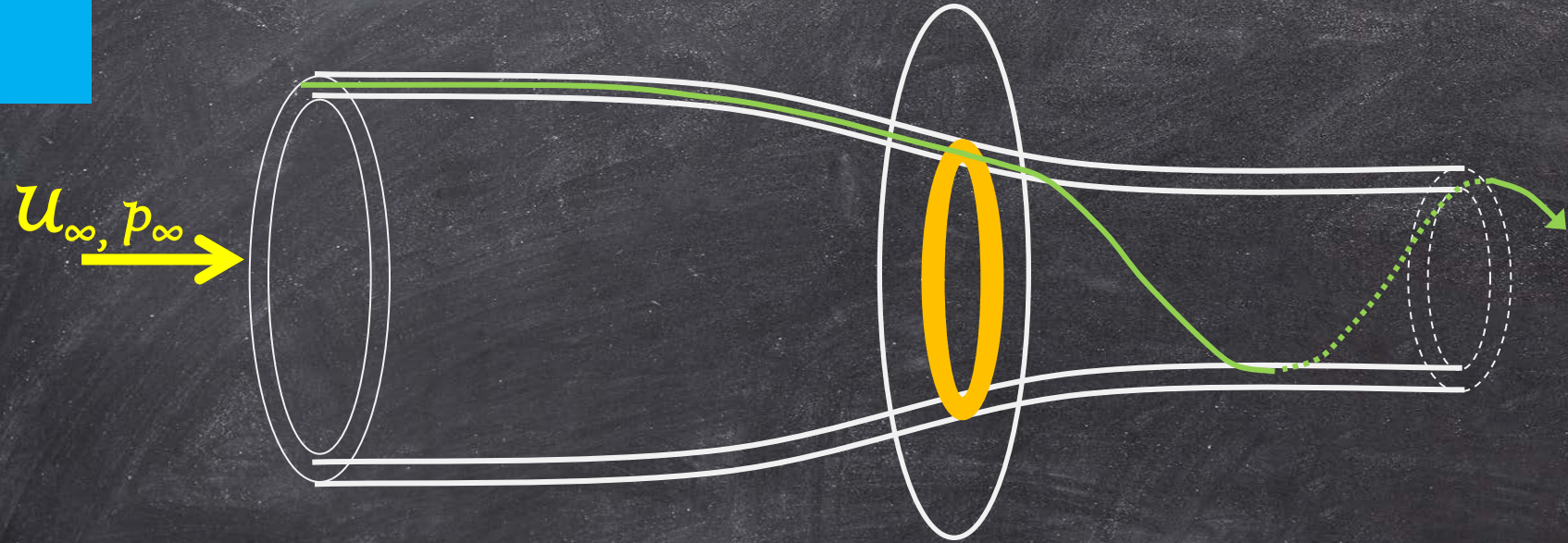
We assume independence of the annuli



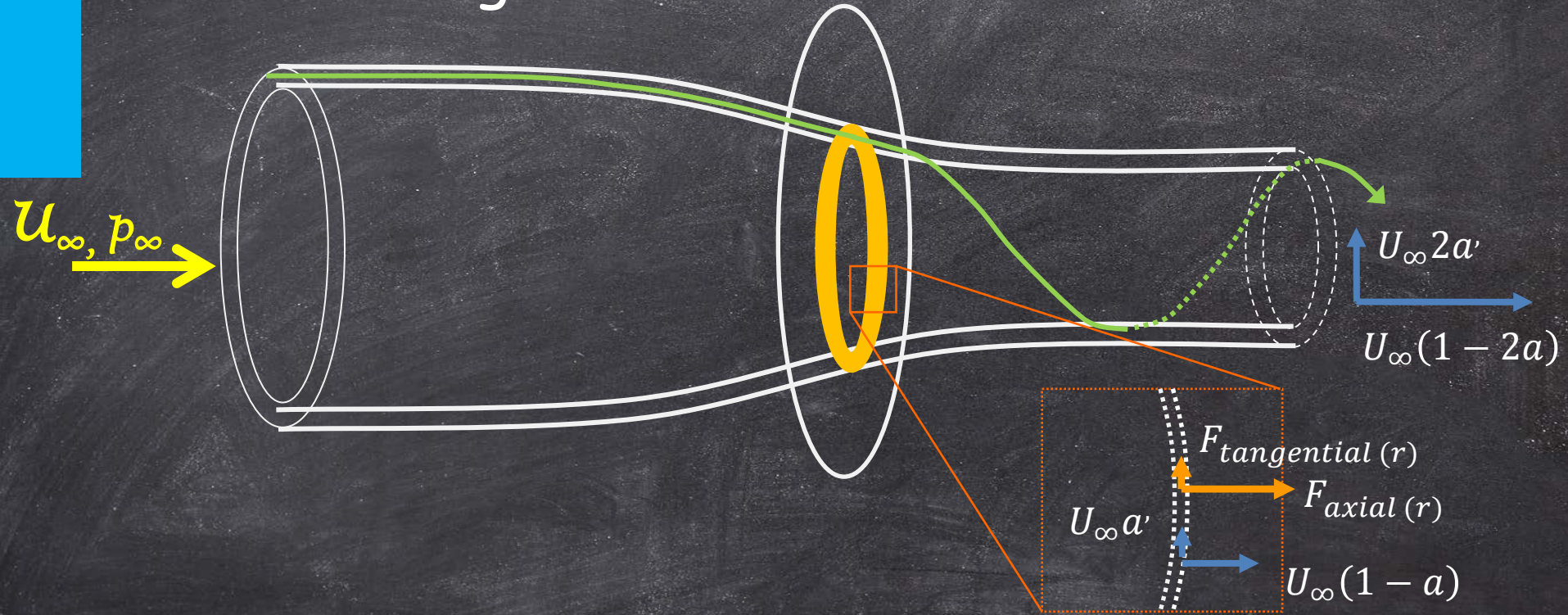
What loading is applied in this streamtube annulus?

- Axial loading, resulting in axial induction
 - Total axial loading will result in thrust
- Tangential loading, resulting in tangential induction
 - Total tangential loading will result in torque

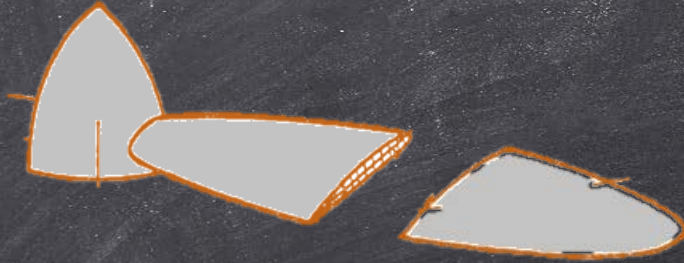
Torque will create a rotation in the wake



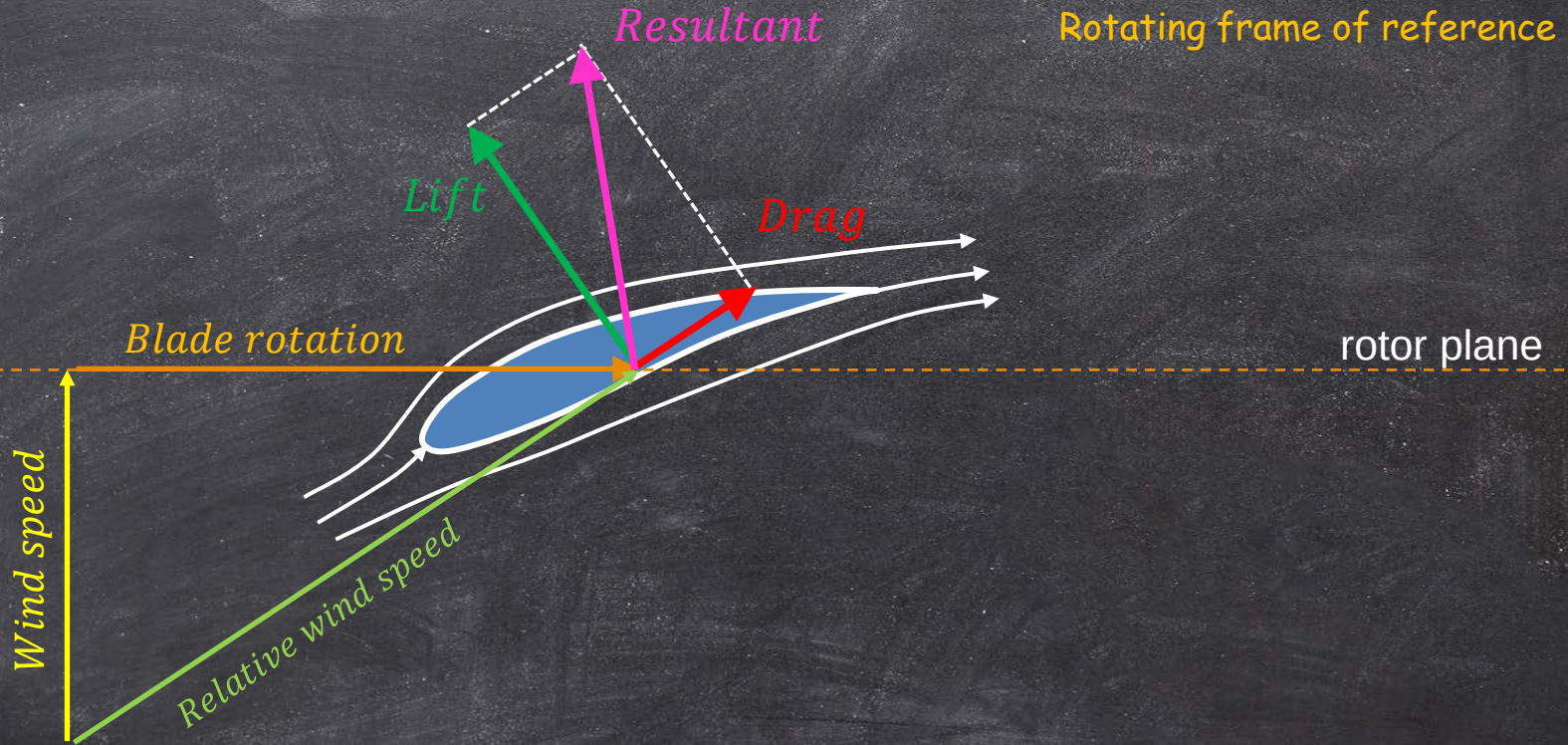
We have at each annulus
a tangential and axial induction



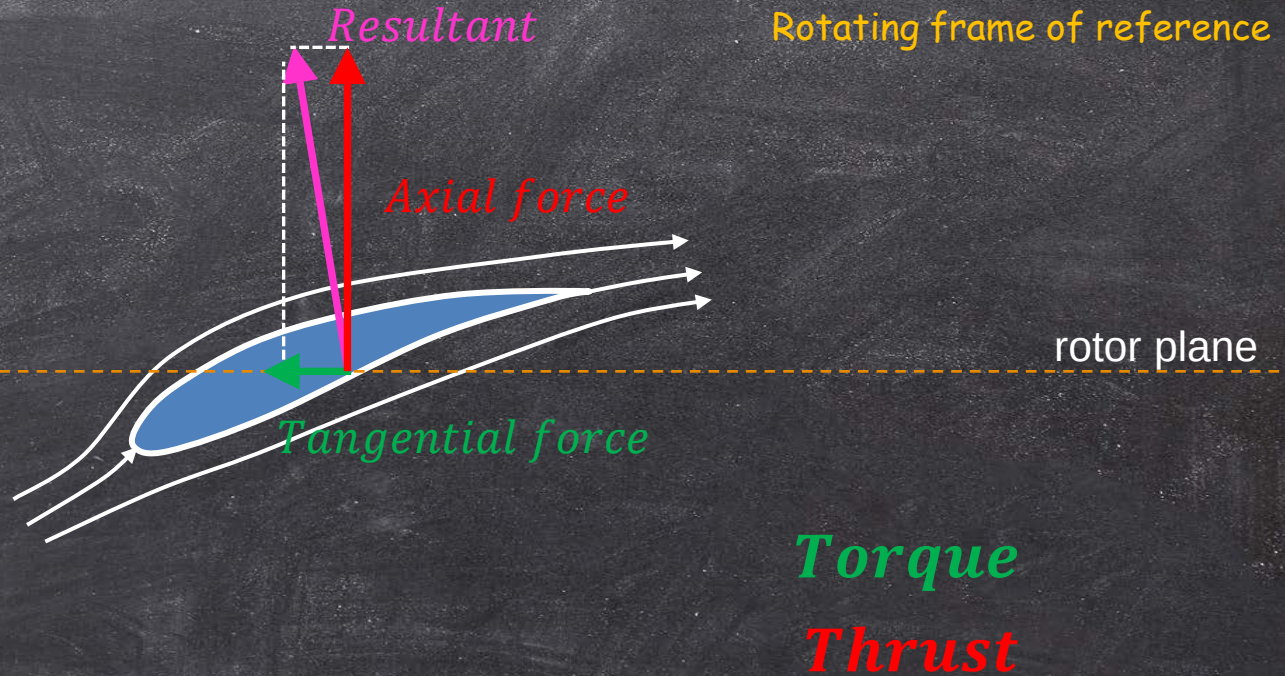
Blade element theory



Velocity triangle and forces

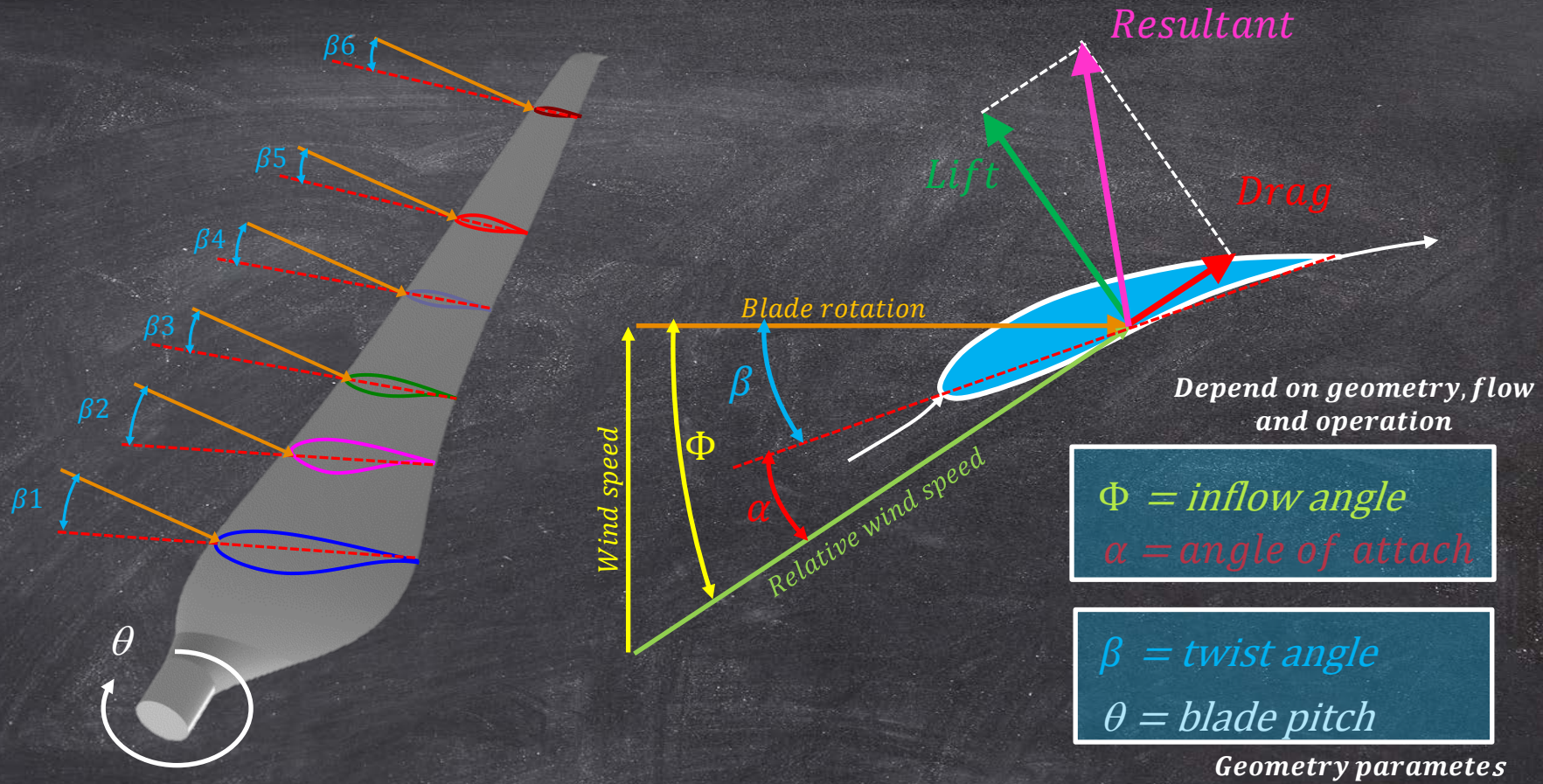


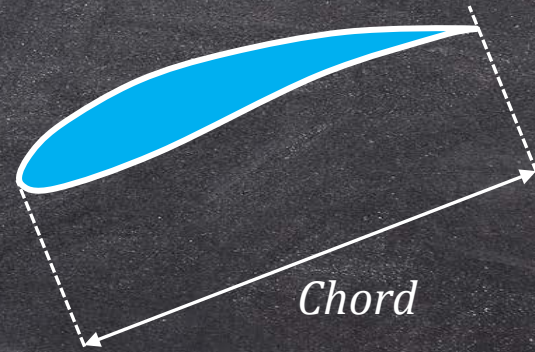
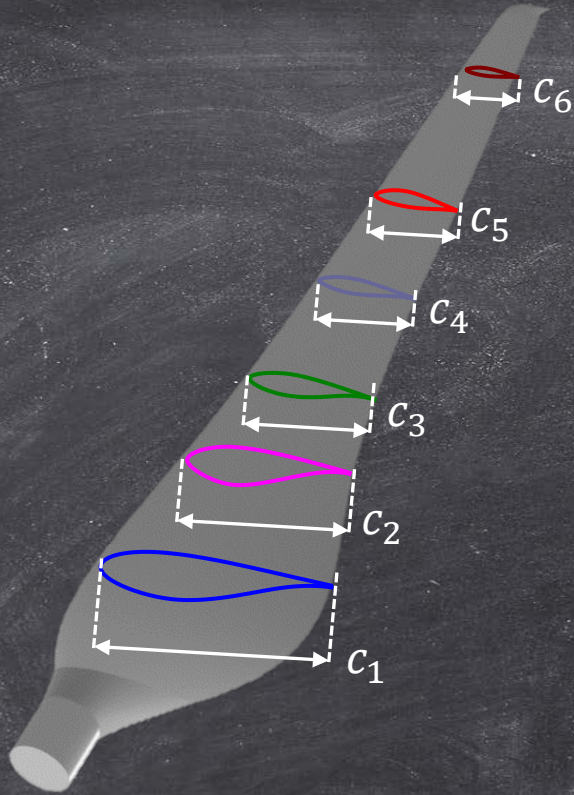
Velocity triangle and forces

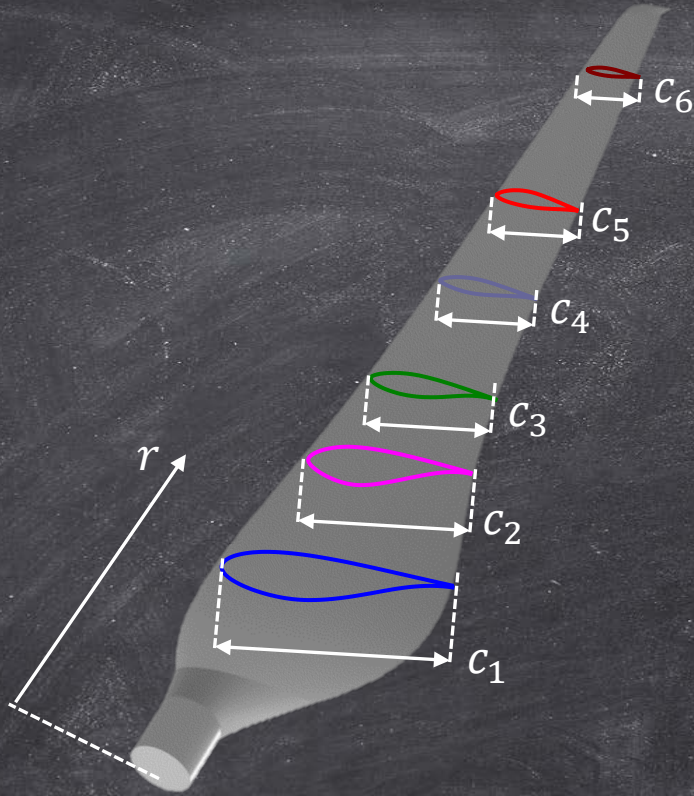


Rotor and Wake Aerodynamics

Rotor design







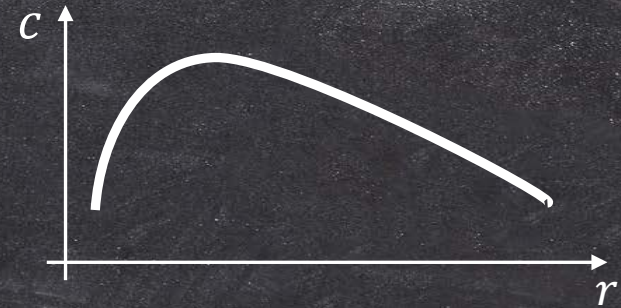
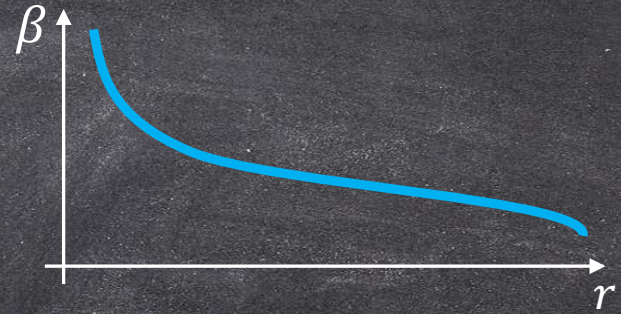
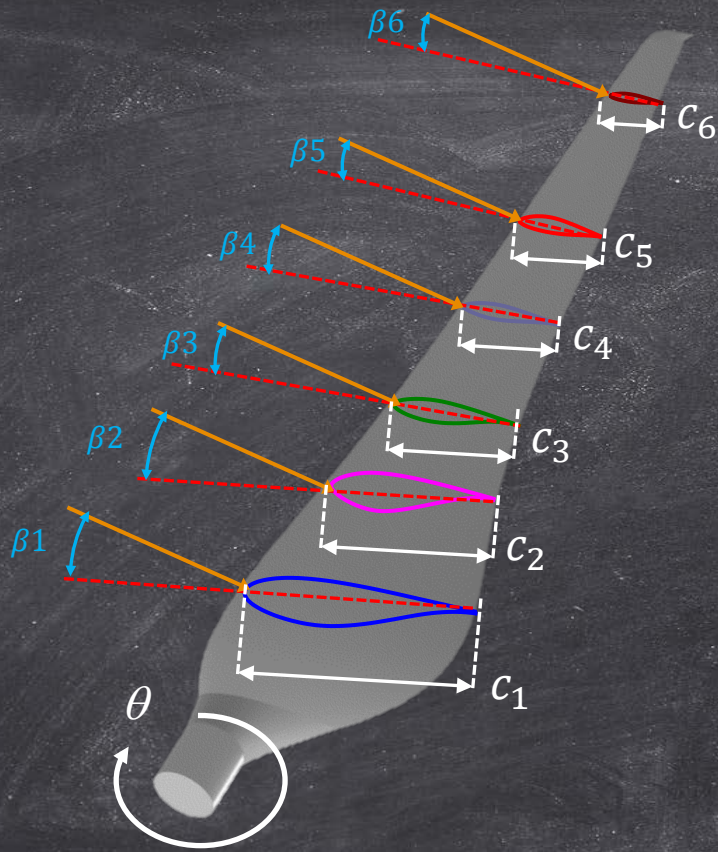
Solidity σ_r

$$\sigma_r = \frac{Nc}{2\pi r}$$

N = number of blades

r = radius

c = chord length





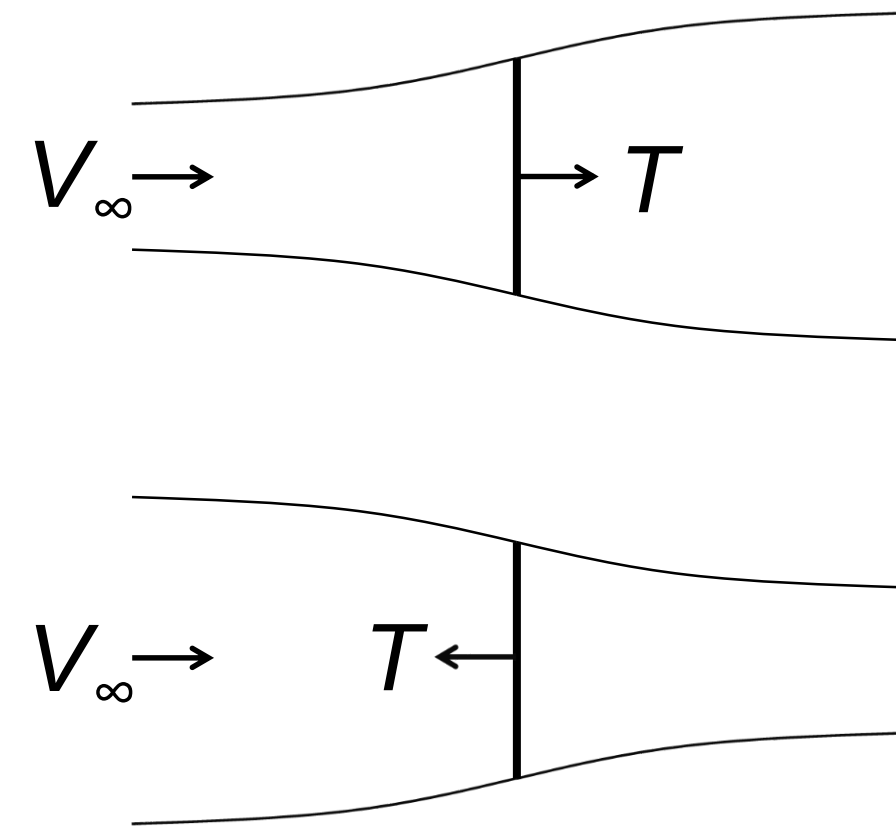
AE4135 Rotor / Wake Aerodynamics

Rotor Performance Coefficients



Rotor Thrust and Power

- Rotors generate thrust and power
- Wind turbines
 - ‘thrust’ force on rotor in freestream direction
 - power *generated*
- Propellers
 - thrust force on rotor opposite to freestream direction
 - power *consumed*
- **What parameters affect the thrust and power, and how?**
 - We will apply dimensional analysis to find out





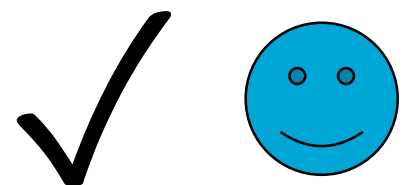
Intermezzo: Dimensional Analysis

- **Dimensional analysis** can provide **functional relations** between physical variables x_i

$$f(x_1, x_2, \dots, x_N) = 0$$

- **Units** within functional relation should be **consistent**
- Example

$$F = m \cdot a \quad \left\{ \begin{array}{l} [F] = \text{N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2} \\ [m] = \text{kg} \\ [a] = \text{m} \cdot \text{s}^{-2} \end{array} \right\} \quad \text{kg} \cdot \text{m} \cdot \text{s}^{-2} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$$





Dimensional Analysis – Buckingham-Pi Theorem

- We distinguish between **basic** and **derived units**
- Basic units
 - Mass $\rightarrow$ kilograms $\rightarrow$ M
 - Length $\rightarrow$ meters $\rightarrow$ L
 - Time $\rightarrow$ seconds $\rightarrow$ T
- Function $f(x_1, x_2, \dots, x_N)$ has N variables with k basic units
 - $\rightarrow$ k variables with independent dimensions
- **Buckingham-Pi Theorem:** express function f into a combination of $N-k$ dimensionless products
 - $\rightarrow f(\Pi_1, \Pi_2, \dots, \Pi_{N-k}) = 0$ (proof omitted here)

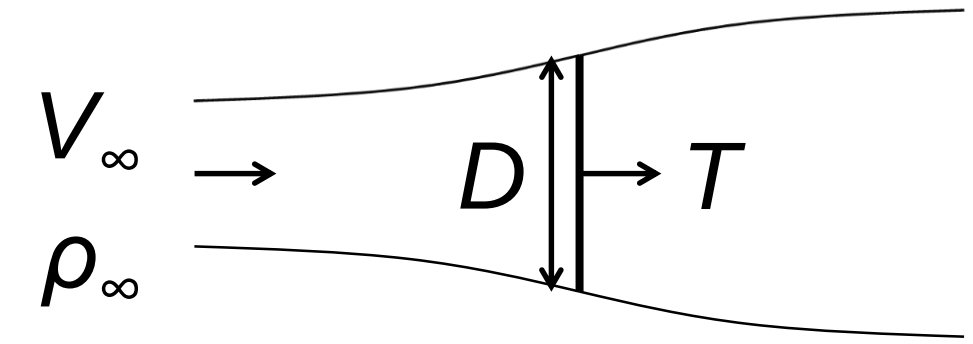


Dimensional Analysis – Pi Products

- Dimensionless products Π_i
 - Combination of k independent vars. and 1 dependent var.:
$$\Pi_i = \left(x_1^{a_{i,1}} \cdot x_2^{a_{i,2}} \cdot \dots \cdot x_k^{a_{i,k}} \right) \cdot x_{k+i}$$
 - Exponents $a_{i,k}$ determined by enforcing consistency of units
- We will now apply the Buckingham-Pi theorem to find the influence of inflow and rotor parameters on
 - Actuator disk thrust
 - Actuator disk power
 - Rotor thrust



Actuator Disk Thrust (1/4)



- Actuator disk thrust depends on inflow conditions and actuator disk size

- Variables involved:

– Thrust	T	$\rightarrow$ N	$\rightarrow$ $\boxed{\text{MLT}^{-2}}$	Dependent
– Inflow momentum	V_∞ (velocity)	$\rightarrow$ m/s	$\rightarrow$ $\boxed{\text{LT}^{-1}}$	Independent
	ρ_∞ (density)	$\rightarrow$ kg/m ³	$\rightarrow$ $\boxed{\text{ML}^{-3}}$	
– Disk size	D (diameter)	$\rightarrow$ m	$\rightarrow$ $\boxed{\text{L}}$	

$$\rightarrow f(T, D, V_\infty, \rho_\infty) = 0$$

- We pick 3 vars with independent dimensions: D , V_∞ , ρ_∞



Actuator Disk Thrust (2/4)

- 3 independent vars. (D , V_∞ , ρ_∞), 1 dependent var. (T)

$$\begin{aligned} [T] &= MLT^{-2} & [V_\infty] &= LT^{-1} \\ [D] &= L & [\rho_\infty] &= ML^{-3} \end{aligned}$$

- We need $4-3 = 1$ Π -product to define f :

$$\begin{aligned} \Pi &= D^a V_\infty^b \rho_\infty^c T = 0 \\ [\Pi] &= [D^a][V_\infty^b][\rho_\infty^c][T] = 1 \end{aligned}$$



Actuator Disk Thrust (3/4)

- We consider consistency of units to find exponents a, b, c

$$[\Pi] = [D^a][V_\infty^b][\rho_\infty^c][T] = 1$$

$$(L)^a(LT^{-1})^b(ML^{-3})^c(MLT^{-2}) = 1$$

- We can define 1 equation per basic unit

$$\left. \begin{array}{l} \text{M:} \quad c + 1 = 0 \\ \text{L:} \quad a + b - 3c + 1 = 0 \\ \text{T:} \quad -b - 2 = 0 \end{array} \right\} \begin{array}{l} a = -2 \\ b = -2 \\ c = -1 \end{array} \Rightarrow \boxed{\Pi = \frac{T}{\rho_\infty V_\infty^2 D^2}}$$

- We can use the Π -product to compare thrust performance of rotors of different scales and in different inflows



Actuator Disk Thrust (4/4)

$$\Pi = \frac{T}{\rho_{\infty} V_{\infty}^2 D^2}$$

- The denominator of the Π -product features terms related to inflow dynamic pressure and rotor disk area
- We multiply the Π -product to make these terms appear explicitly, and obtain the thrust coefficient

$$C_T = \frac{T}{\frac{1}{2} \rho_{\infty} V_{\infty}^2 \frac{\pi}{4} D^2} = \frac{T}{q_{\infty} A_{\text{disk}}}$$

(for propellers, this coefficient is usually denoted T_c)



Actuator Disk Power

- We can take the same approach to find the relation between actuator disk power, disk size, and inflow conditions

[Pause to do the derivation yourself]

$$[\Pi] = [D^a][V_\infty^b][\rho_\infty^c][P] = 1$$

$$(L)^a (LT^{-1})^b (ML^{-3})^c (ML^2T^{-3}) = 1$$

$$\Rightarrow \boxed{\Pi = \frac{P}{\rho_\infty V_\infty^3 D^2}}$$

- This Π -product is used to define the power coefficient

$$C_P = \frac{P}{\frac{1}{2} \rho_\infty V_\infty^3 \frac{\pi}{4} D^2} = \frac{P}{\frac{1}{2} \rho_\infty V_\infty^3 A_{\text{disk}}} \quad \begin{array}{l} \text{(for propellers, this} \\ \text{coefficient is usually} \\ \text{denoted } P_C) \end{array}$$



Rotor Thrust (1/2)

- We can extend the analysis to the case of a rotor
- The thrust of the rotor depends on 1 additional parameter compared to the case for the actuator disk
 - Rotor size D (diameter)
 - Inflow momentum V_∞ (inflow velocity)
 - Inflow momentum ρ_∞ (air density)
 - Rotor operating conditions n (rotational speed)
- We now need $5-3 = 2$ Π -products to define f

} Same as before

$$\Pi_1 = D^a V_\infty^b \rho_\infty^c T = 0$$

$$\Pi_2 = D^a V_\infty^b \rho_\infty^c n = 0$$



Rotor Thrust (2/2)

- *[Pause to do the derivation yourself]*
 - $\Pi_1 = \frac{T}{\rho_\infty V_\infty^2 D^2} \rightarrow$ same **thrust coefficient** as before
 - $\Pi_2 = \frac{nD}{V_\infty} \rightarrow$ **tip-speed ratio**, typically written as $\lambda = \frac{\Omega R}{V_\infty}$ for wind turbines
- We can also take n as independent variable instead of V_∞
[Pause to do the derivation yourself]
 - $\Pi_1 = \frac{T}{\rho_\infty n^2 D^4} \rightarrow$ alternative **thrust coefficient**, typically used for **isolated propellers** and also called C_T
 - $\Pi_2 = \frac{nD}{V_\infty} \rightarrow$ alternative tip-speed ratio, inverse is typically used for propellers and called **advance ratio** (J)



In this module..

- We introduced the concept of dimensional analysis
- We used dimensional analysis to define performance coefficients for an actuator disk and a rotor to describe the effects of rotor size and operating conditions on the thrust and power
- The performance coefficients can be computed using numerical methods, as discussed in later modules